

Hardware Implementation Of Four Degree Of Freedom Deep Reinforcement Learning Arm Control Model Trained Through Human Demonstrations

Roel Jose Hernandez Chaudary hernaroe@sonoma.edu

Advisor: Sudhir Shrestha

Abstract

- The purpose of this study was to explore the ability to control a 4-DOF Robot Arm by implementing a trained model through human demonstration.
- Further, to determine if it is possible to control a different model of 4-DOF robot arm than the one that was used to train the model.

Background

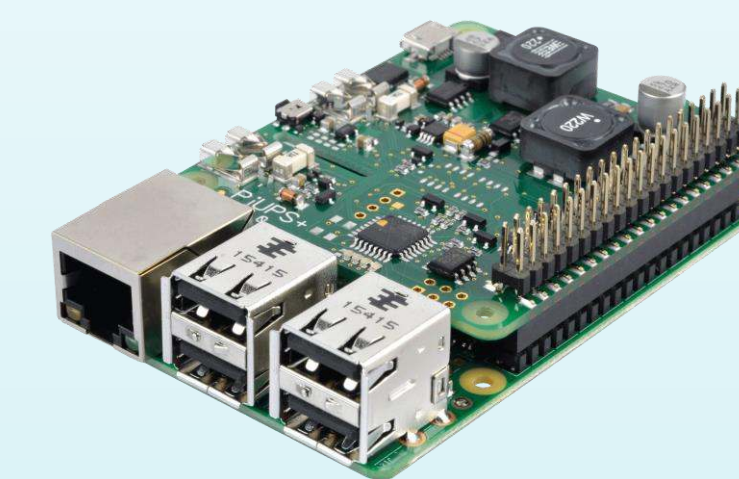
- Robotics have been adopted by many companies to assist humans in applications that are tedious, labor intensive, or dangerous.
- For a given application, resources are spent on acquiring a robot and a robotic engineer to develop a solution for that application and configuration.
- Rather than having a robotic engineer hard code the operation of a robot for a new application, companies could utilize deep reinforcement learning to train an intelligent agent to operate a robot for that specific application.
- This would allow resources companies invest in robotics to be more robust for different applications and purposes.

Implementation

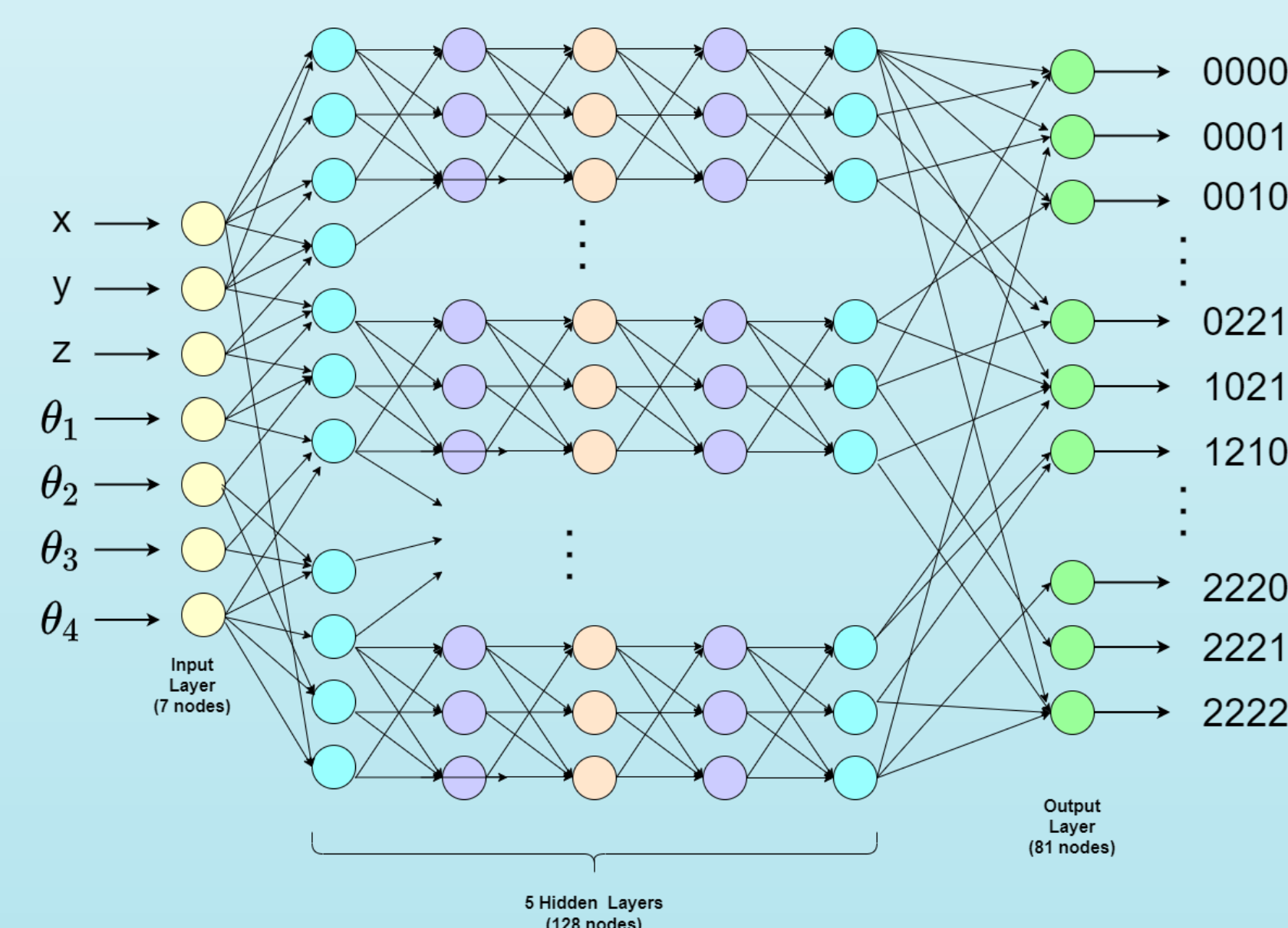
Dobot Magician
Robot Arm



Raspberry Pi



Neural Network Configuration



Results

	ArbotiX	Dobot
Average Distance	3.051 cm	6.155 cm
Episodes < 5 cm	83%	27%
Episodes < 1 cm	18%	0%

Conclusions

- The implementation showed that the simulation of the Trained model behave well when used on the same type of robot (PhantomX Robot Arm),
- When using a different 4-DOF Robot Arm the behavior was not similar, even when we were able to implement the trained model on Dobot Magician Robot Arm, the accuracy to reach the target was not as expected.

Future work

- We can implement our trained model in different robot arm making the adjustment in the DH parameters to compare the accuracy to reach the target position.
- The accuracy is expected to improve by using a Robot Arm that has similar characteristic to ArbotiX Robot Arm.
- Training the model with a bigger set of data should achieve better behavior and accuracy in the implementation.